# MILITARY SPECIFICATION

## CONVERTER, LIQUID OXYGEN GCU-11/A

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers one type of liquid oxygen converter with a capacitance gaging element. The converter is designated Type GCU-11/A. The converter includes warming coils.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-V-9050 | Valve, Oxygen, Pressure Relief, Aircraft |
| MIL-C-25516 | Connectors, Electrical, Miniature, Environmental Resistant Type, General Specification For |
| MIL-C-25666 | Converters, Liquid Oxygen, Capacitance Type Gaging, General Specification for |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-831 | Test Reports, Preparation of |

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General specification. The requirements of MIL-C-25666 shall apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.2 Qualification. Converters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.3 Design. The converter shall conform to the dimensions and general design shown on figures 1 and 2. The converter shall have a minimum capacity of 10 liters of liquid oxygen measured at 1 atmosphere, plus at least 3 percent of this volume to allow for the change in liquid density at the operating pressure. The converter shall be designed for an operating pressure of 300 psig.

3.3.1 Schematic diagram. The converter shall be designed in accordance with the schematic diagram shown in figure 3.

3.3.2 Design strength. The filled converter shall be designed to withstand the following "g" forces without damage:

| | | |
|---|---|---|
| Vertical | (down) | 10g |
| Vertical | (up) | 5g |
| Lateral | | 5g |

3.3.3 Weight. The weight of the complete converter shall not exceed 40 pounds.

3.4 Performance.

3.4.1 Final assembly pressure. The pressure used for the final assembly pressure test shall be 320 psig.

3.4.2 Filling time. Filling time shall not exceed 5 minutes when the converter is tested as specified.

3.4.3 Evaporation loss. The evaporation loss of the converter shall not exceed 3 pounds of oxygen when the converter is tested as specified.

3.4.4 Buildup time. Buildup time shall not exceed 10 minutes when the converter is tested as specified.

3.4.5 Capacitance. The capacitance of the sensing element, after it is installed in the empty converter, shall be 123.5 (plus or minus 1.0) picofarad (pF).



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3.4.6 Permanent volumetric expansion. The permanent volumetric expansion of the inner vessel shall not exceed 10 percent of the total volumetric expansion while the inner vessel is subject to a pneumatic or hydrostatic expansion pressure of 500 psig.

3.4.7 Delivery rate. The converter shall deliver constant gas flows up to 100 liters of gaseous oxygen per minute for not less than 1 hour with the operating pressure remaining between 295 and 315 psig after the first 15 minutes of operation. An intermittent increase or decrease of pressure by an additional 5 psig is allowable provided these variances do not exceed 2-minutes of operation, the temperature of the gaseous oxygen shall not fall below 40 degrees Fahrenheit below ambient temperature.

3.4.8 Calibration. The capacitance sensing element shall have characteristics that provide for a linear relationship between electrical capacitance and volume of liquid oxygen sensed. The relationship shall be in accordance with figure 4, using the equations that are specified in 3.4.8.1 to determine tolerances. An example of use of the equations is as follows: At the 6-liter test point, suppose the actual quanitity of liquid oxygen is 6.1 liters. The measured capacitance (C) must be between C maximum (max) and C minimum (min):

C max = 5.9 x 6.1 + 124.7 = 160.7 Picofarad (pF)

C min = 5.7 x 6.1 + 122.3 = 157.3 pF

3.4.8.1 Equations. The following equations shall be used to calculate the maximum and the minimum capacitance.

| Values less than 2 liters | Values greater than 2 liters |
|---|---|
| C minimum = 5.6L + 122.5 | C minimum = 5.7L + 122.3 |
| C maximum = 6L + 124.5 | C maximum = 5.9L + 124.7 |

C = Capacitance in pF (Picofarad)
L = Exact quantity in liters of liquid oxygen at test point

3.4.9 Operating temperature. The operating pressure shall remain between 295 and 320 psig after the first 15 minutes of flow at the low temperature and shall remain between 290 and 315 psig after the first 15 minutes of flow at the high temperature. An intermittent increase or decrease of pressure by an additional 5 psig is allowable provided these variances do not exceed a 2-minute interval for any one pressure cycle.

3.4.10 Vibration. The converter shall be vibrated in accordance with MIL-C-25666.

3.5 Pressure relief valves. The pressure relief valves shall conform to types III and IV of MIL-V-9050 and shall be of such size and shape to fit the over-all dimensions specified for the converter.

3.6 Marking of ports. Ports shall be marked as indicated in figure 1.

4. QUALITY ASSURANCE PROVISIONS



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4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Classification of inspection. The inspection required by this specification is classified as follows:

(a) Quality conformance inspection
(b) Qualification inspection

4.3 Quality conformance inspection. The quality conformance inspection shall be in accordance with MIL-C-25666.

4.4 Qualification inspection. The qualification inspection shall consist of all of the methods of inspections that are described in MIL-C-25666.

4.4.1 Samples. At least two converters shall be subjected to the qualification inspection. One converter shall be submitted to the qualifying activity (see 6.3). The converter that is submitted to the qualifying activity may be a tested converter provided it has not been subjected to the gunfire test that is described in MIL-C-25666.

4.4.2 Report. Three copies of a qualification inspection report that has been prepared in accordance with MIL-STD-831 shall be submitted to the qualifying activity.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with MIL-C-25666 (see 6.2).

6. NOTES

6.1 Intended use. The converter covered by this specification is intended to be used in the F-101B aircraft to provide storage and pressurization of 10 liters of liquid oxygen to 300 psig nominal pressure.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Selection of applicable levels of preservation and packaging and packing of MIL-C-25666 (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List (QPL-25972) whether or not such products have actually been so listed by the date. The attention of the contractor is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible



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to be awarded contracts or orders for the products covered bt this specification. The activity responsible for this Qualified Products List is OC-ALC/MMEDO, Tinker Air Force Base OK 73145, and information pertaining to qualification of products may be obtained from that activity.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force-99

Reviewer:
Air Force-11

Preparing activity:
Air Force-71

Project No.
1660-F484



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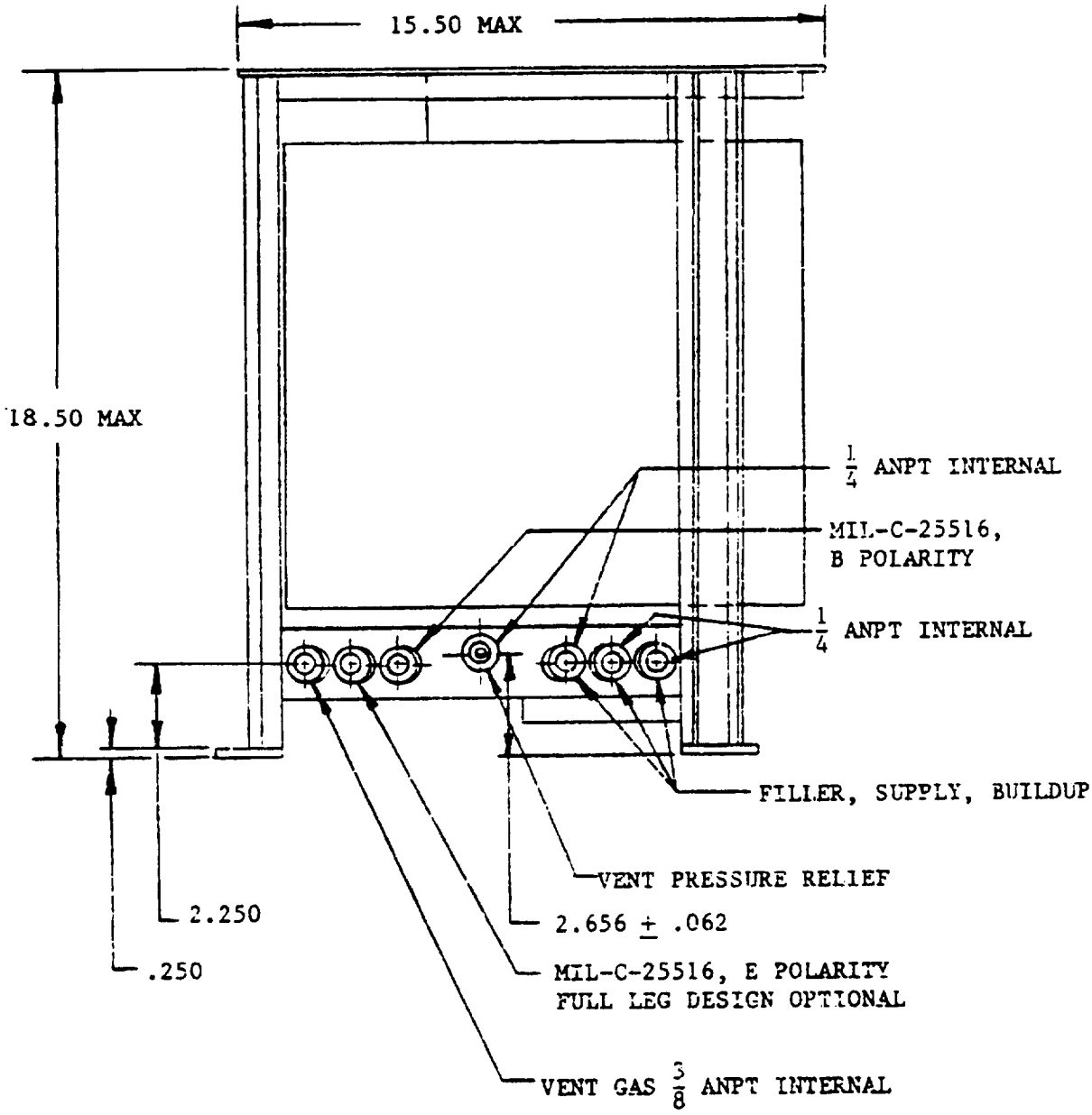
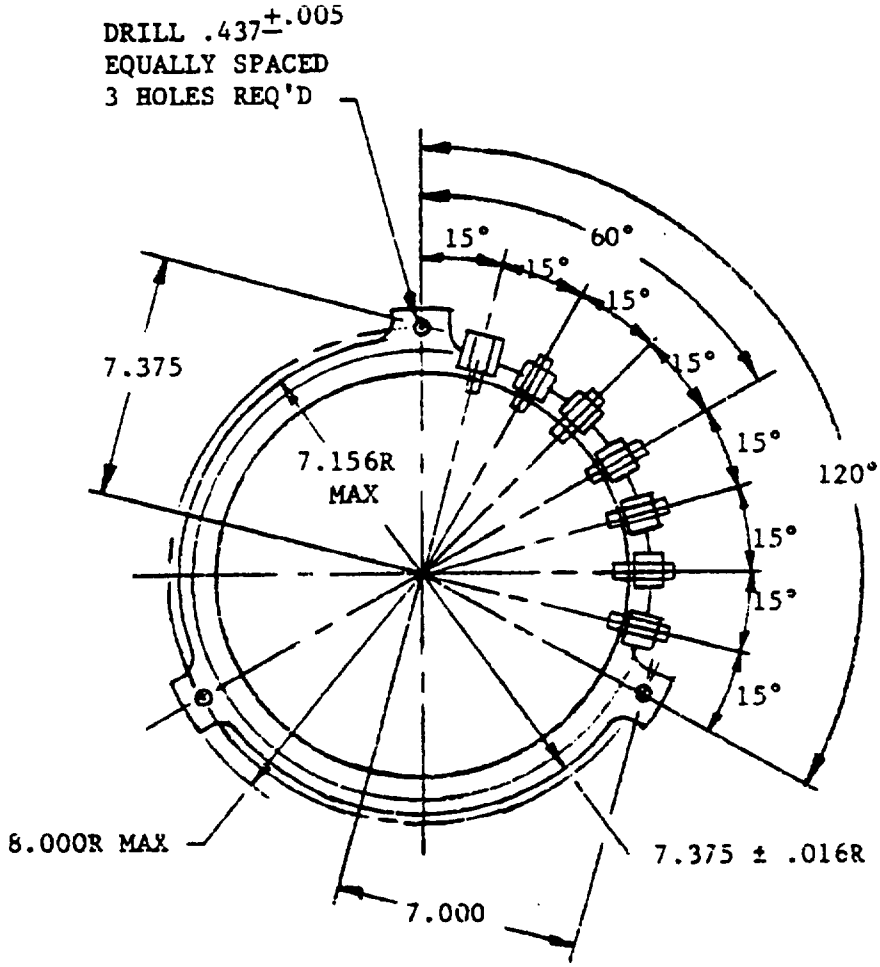



Figure 1. General design of 10-liter converter.



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TYPE OF MOUNTING BASE OPTIONAL TO CONTRACTOR, EITHER THREE MOUNTING LEGS OR A MOUNTING RING.

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.
TOLERANCES ON: DECIMALS ± .062 ANGLES ± 1°

Figure 2. Bottom view of converter.



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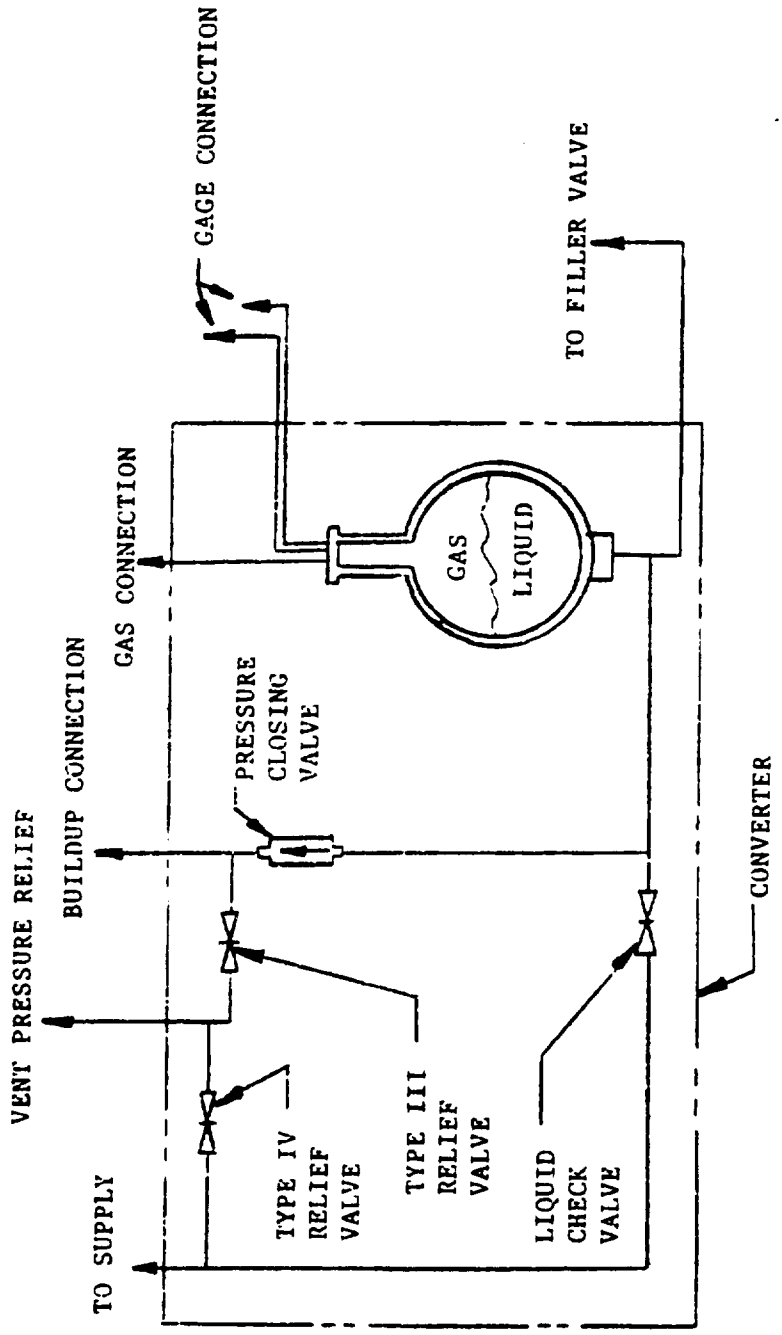
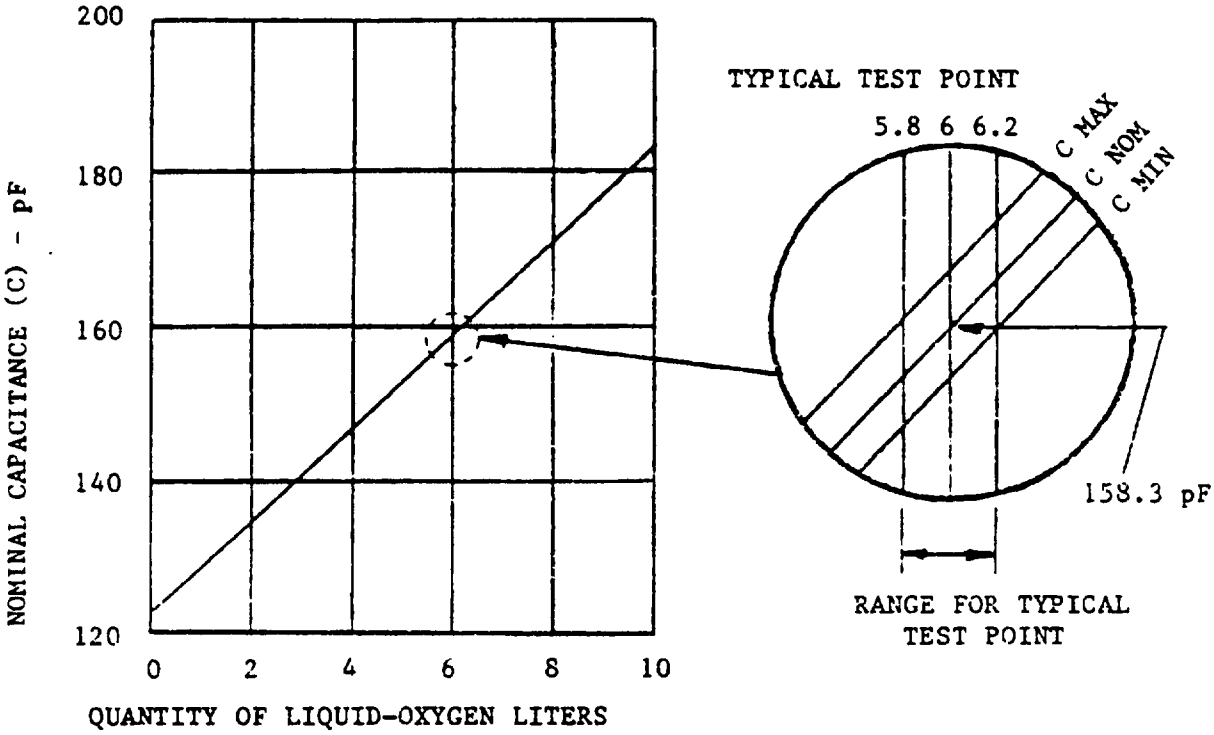



Figure 3. Schematic diagram of converter with capacitance gaging element.



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| LIQUID OXYGEN (LITERS) | NOMINAL CAPACITANCE (pF) |
|---|---|
| 0 | 123.5 |
| 2 | 135.1 |
| 4 | 146.7 |
| 6 | 158.3 |
| 8 | 169.9 |
| 10 | 181.5 |

*Figure 4. Relationship between capacitance (C) and volume of oxygen sensed.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-25972C(USAF) | CONVERTER, LIQUID OXYGEN GCU-11/A |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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